Ask

WEBHOOKS

Ask

Overview

Subscribe to real-time event notifications from Fleet Management platform

Webhooks allow your application to receive real-time notifications when events occur in the Fleet Management system. Instead of polling for updates, the platform will send HTTP POST requests to your configured endpoint whenever subscribed events happen.

What are Webhooks?

Webhooks are HTTP callbacks that deliver event data to your server in real-time. When an event you've subscribed to occurs (such as a training activity starting or ending), the Fleet Management platform will send a POST request to your webhook URL with the event details.

Available Events

Currently supported event types:

- skills-activity-started - Triggered when a learner begins a training activity
- skills-activity-ended - Triggered when a learner completes or exits a training activity

Additional event types may be added in the future as the platform evolves.

Registering a Webhook

Webhooks are managed through the Fleet Management application at the organization level.

Prerequisites

- Admin access to your organization in the Fleet Management application
- An HTTPS endpoint capable of receiving POST requests
- Ability to verify HMAC-SHA256 signatures

Registration Steps

1. Navigate to your Organization Settings in the Fleet Management application
2. Go to the **Webhooks** section
3. Click **Add Webhook**
4. Configure your webhook:
   - **URL**: Your HTTPS endpoint (e.g., `https://your-domain.com/webhooks/fleet-management`)
   - **Event Types**: Select which events to subscribe to
   - **Description**: Optional note for your reference
5. Click **Save**
6. **Important**: Copy the **signing secret** shown - it will only be displayed once!

Managing Webhooks

You can manage your webhooks through the Fleet Management application:

- **Enable/Disable**: Toggle webhooks on or off without deleting
- **Update URL**: Change the destination endpoint
- **Change Event Types**: Subscribe or unsubscribe from specific events
- **Regenerate Secret**: Generate a new signing secret (invalidates the old one)
- **Delete**: Permanently remove a webhook configuration

Webhook Security (HMAC-SHA256)

All webhook requests are signed using HMAC-SHA256 to ensure authenticity. Your application **must verify** the signature to confirm requests are genuinely from Fleet Management.

Signature Verification

Each webhook request includes these headers:

```
X-Fleet-Signature-256: <hex-signature>
X-Fleet-Timestamp: <unix-timestamp>
X-Fleet-Request-Id: <request-id>
```

Verification Steps:

1. Extract the timestamp and signature from headers
2. Construct the signed content: `{timestamp}.{raw-request-body}`
3. Compute HMAC-SHA256 using your signing secret
4. Compare the computed signature with the received signature using constant-time comparison
5. Optionally: reject requests with timestamps older than 5 minutes (replay protection)

Example Code

C# | Python | Node.js

```
using System.Security.Cryptography;
using System.Text;

public bool VerifyWebhook(string signingSecret, string body,
HttpRequest request)
{
    var timestamp = request.Headers["X-Fleet-Timestamp"];
    var signature = request.Headers["X-Fleet-Signature-256"];

    var signedContent = $"{timestamp}.{body}";
    var secretBytes =
Convert.FromBase64String(signingSecret);

    using var hmac = new HMACSHA256(secretBytes);
    var hashBytes =
hmac.ComputeHash(Encoding.UTF8.GetBytes(signedContent));
    var expectedSignature =
Convert.ToHexString(hashBytes).ToLowerInvariant();

    return CryptographicOperations.FixedTimeEquals(
        Encoding.UTF8.GetBytes(expectedSignature),
        Encoding.UTF8.GetBytes(signature));
}
```

Webhook Payload Structure

All webhook payloads follow this general structure:

```
{
  "requestId": "98AAF1CE-3AA1-F011-B383-
6045BD4A3A04|638969523968650000|a1afe94510cf36dd",
  "eventType": "skills-activity-started",
  "timestamp": "2026-01-08T14:30:00Z",
  "stationId": "507f1f77bcf86cd799439011",
  "data": {
    // Event-specific fields (varies by event type)
  }
}
```


Common Fields:

- `requestId` - Unique request identifier from the original event (use for idempotency)
- `eventType` - Type of event that occurred
- `timestamp` - ISO 8601 UTC timestamp when the event occurred
- `stationId` - Fleet Management station identifier
- `data` - Event-specific payload (structure varies by event type)

See individual event type pages for detailed payload structures.

Retry Logic

If your endpoint returns a non-2xx response or times out, the Fleet Management platform will automatically retry delivery with exponential backoff:

Attempt	Delay	Total Elapsed
1	Immediate	0
2	1 minute	1 minute
3	5 minutes	6 minutes
4	15 minutes	21 minutes
5	1 hour	1h 21m
6	6 hours	7h 21m

After 6 failed attempts, delivery is abandoned.

Success Criteria

A webhook delivery is considered successful when:

- Your endpoint returns HTTP status 2xx (200-299)
- Response is received within 30 seconds

Best Practices

- **Respond quickly**: Return 2xx immediately and process asynchronously
- **Handle duplicates**: Use `requestId` or `X-Fleet-Request-Id` for idempotency
- **Validate signatures**: Always verify HMAC-SHA256 before processing
- **Log failures**: Track webhook processing errors for debugging
- **Use HTTPS**: Required for production webhooks

Troubleshooting

Common Issues

Signature verification fails

- Ensure you're using the raw request body (no parsing)
- Verify signing secret is correct (check Admin Portal)
- Confirm you're base64-decoding the secret before HMAC computation

Missing events

- Check webhook is enabled in Admin Portal
- Verify event types are subscribed
- Check your server logs for delivery failures
- Review Hangfire dashboard (ask Laerdal support for access)

Timeout errors

- Ensure your endpoint responds within 30 seconds
- Process webhooks asynchronously if needed
- Check for network/firewall issues

Need Help?

Contact Laerdal support for assistance with:

- Webhook configuration issues
- Delivery failure investigation

- Access to monitoring dashboards
- Custom event type requests

Previous
Authentication

Next
skills-activity-started

Last updated 4 months ago